

[user]

I'm making an ML-based audio "cough detector" algorithm. In this type of algorithm, what pipeline(s) of audio preprocessing are recommended? I assume there's something more specialized than just FFT.

[assistant]

You're right—most cough-detection systems use a **more specialized audio pipeline than a raw FFT**, borrowing heavily from speech and bioacoustics. Below is a **practical, ML-oriented breakdown** of what's commonly used *and why*, with options depending on whether you're doing classical ML or deep learning.

1. Typical High-Level Pipeline

Audio → Preprocessing → Time-frequency representation → Features / Model → Post-processing

For cough detection, the goal is to capture:

- **Transient, explosive onsets**
 - **Broadband noise bursts**
 - **Characteristic temporal structure (inspiratory + expiratory phases)**
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2. Audio Preprocessing (Critical for Coughs)

2.1 Sampling & Channel

- **Sample rate:** 16 kHz (most common), sometimes 22.05 kHz
- **Mono**
- **Bit depth:** 16-bit PCM

Coughs are mostly < 8 kHz, so higher rates bring diminishing returns.

2.2 Bandpass Filtering

Cough energy is concentrated roughly in:

100 Hz – 6 kHz

Common filter:

- High-pass: 80–120 Hz (remove rumble)
- Low-pass: 6–8 kHz (remove irrelevant HF noise)

This alone improves SNR significantly.

2.3 Amplitude Normalization

- Peak normalization or RMS normalization
- Optional **per-window energy normalization**

Avoid aggressive AGC—cough loudness is informative.

2.4 Silence Removal / Activity Detection

You almost never want to classify long silent regions.

Options:

- Energy-based VAD
- WebRTC VAD
- Learned VAD model

Coughs are **short, impulsive**, so this step is extremely helpful.

3. Time–Frequency Representations (Better Than Raw FFT)

3.1 Mel Spectrogram (Most Common)

The gold standard for cough detection.

Typical parameters:

- Window: 20–40 ms
- Hop: 10 ms
- FFT size: 512 or 1024
- Mel bins: 40–128
- Log scaling (log-mel)

Why it works:

- Compresses irrelevant spectral detail
- Emphasizes perceptual bands
- Works well with CNNs

📌 **Most published cough-detection papers use log-mel spectrograms**

3.2 MFCCs (Classical ML Friendly)

If using SVM, RF, GMM, etc.

Typical:

- 13–40 MFCCs
 - Δ and $\Delta\Delta$
- Mean / variance pooling over segments

Pros:

- Compact
- Robust

Cons:

- Can lose temporal microstructure of coughs
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3.3 Constant-Q Transform (CQT)

Less common, but sometimes useful:

- Better low-frequency resolution
- More musically motivated than physiologically motivated

Generally inferior to mel for cough detection.

3.4 Raw Waveform (Advanced DL)

Possible, but:

- Requires lots of data
- Sensitive to noise
- Harder to generalize

Used in models like:

- WaveNet-style CNNs
 - 1D CNN + attention
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4. Specialized Features for Cough Detection

Many systems add **non-speech features** on top of mel/MFCC:

4.1 Temporal / Energy Features

- Short-time energy
- Zero-crossing rate
- Crest factor
- Attack time
- Burst duration

Coughs have:

- Sharp onset
 - High energy burst
 - Short duration (200 ms – 1 s)
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4.2 Spectral Shape Features

- Spectral centroid
- Spectral rolloff
- Spectral flatness (coughs are noise-like)
- Spectral entropy

4.3 Modulation Features

- Energy modulation spectrum
- Temporal envelopes

Useful to distinguish coughs from:

- Speech plosives
- Throat clearing
- Sneezes

5. Deep Learning Architectures (Matched to Features)

5.1 CNN on Log-Mel Spectrograms (Most Proven)

Audio → Log-Mel → CNN → Dense → Cough / Non-cough

Variants:

- 2D CNN
- CNN + LSTM / GRU
- CNN + temporal attention

5.2 CRNN (Very Popular)

- CNN extracts spectral patterns
- RNN captures cough temporal structure

5.3 Event Detection (Not Just Classification)

If you want cough timestamps:

- Sliding window classification

- Post-processing with:
 - Median filtering
 - Hysteresis thresholds
 - Minimum duration constraints
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6. Post-Processing (Often Overlooked)

Coughs are **events**, not frames.

Common rules:

- Merge detections within 200–500 ms
- Reject events shorter than ~100 ms
- Enforce refractory period between coughs

This dramatically reduces false positives.

7. Recommended Baseline Pipeline (Proven & Practical)

If you want a **strong starting point**:

Audio (16 kHz mono)
→ Bandpass (100–6000 Hz)
→ RMS normalization
→ VAD
→ Log-Mel Spectrogram (64–128 bins)
→ CNN or CRNN
→ Event-level post-processing

This is very close to what's used in:

- COVID cough detectors
 - Respiratory sound analysis systems
 - Mobile health cough counters
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8. Dataset Quality Matters More Than Features

A mediocre model + good cough labels **beats** a great model + noisy labels.

Key dataset issues:

- Microphone variability
 - Background speech
 - Throat clears mislabeled as coughs
 - Class imbalance (huge)
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